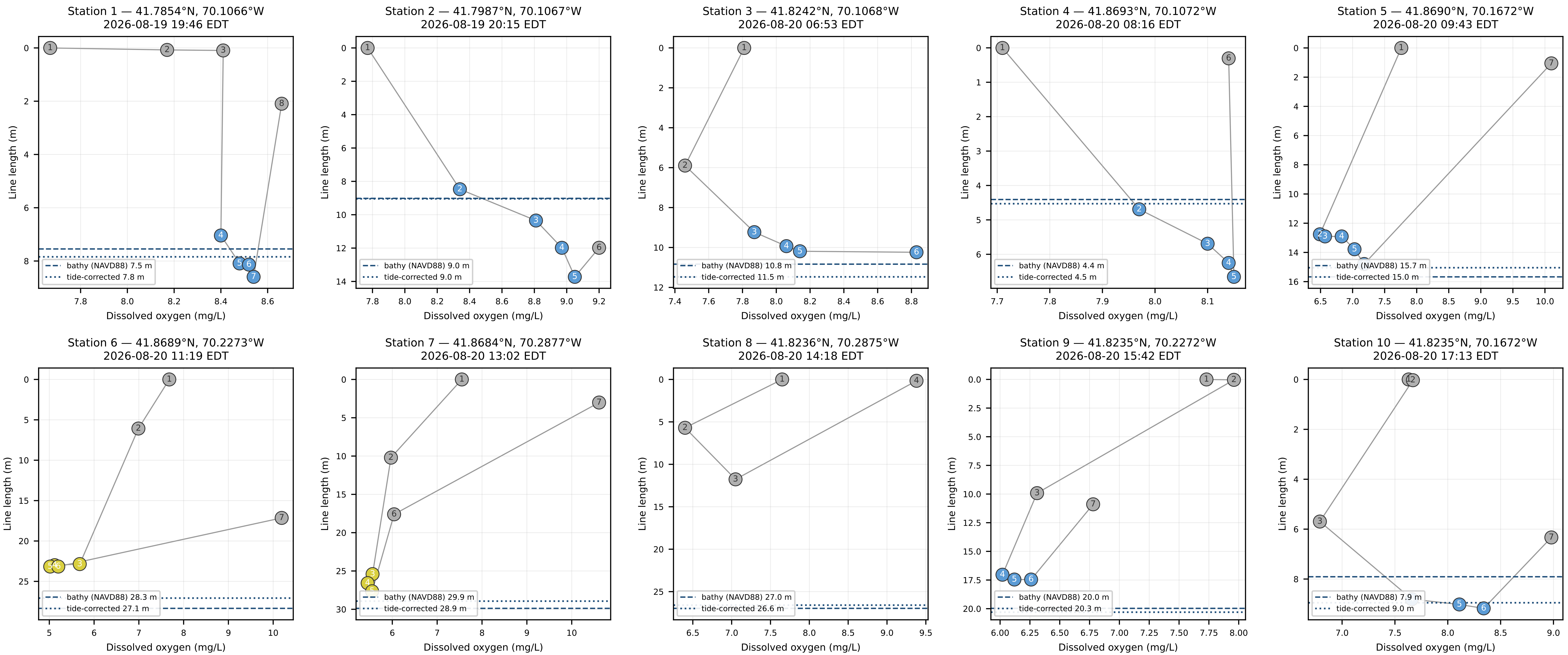
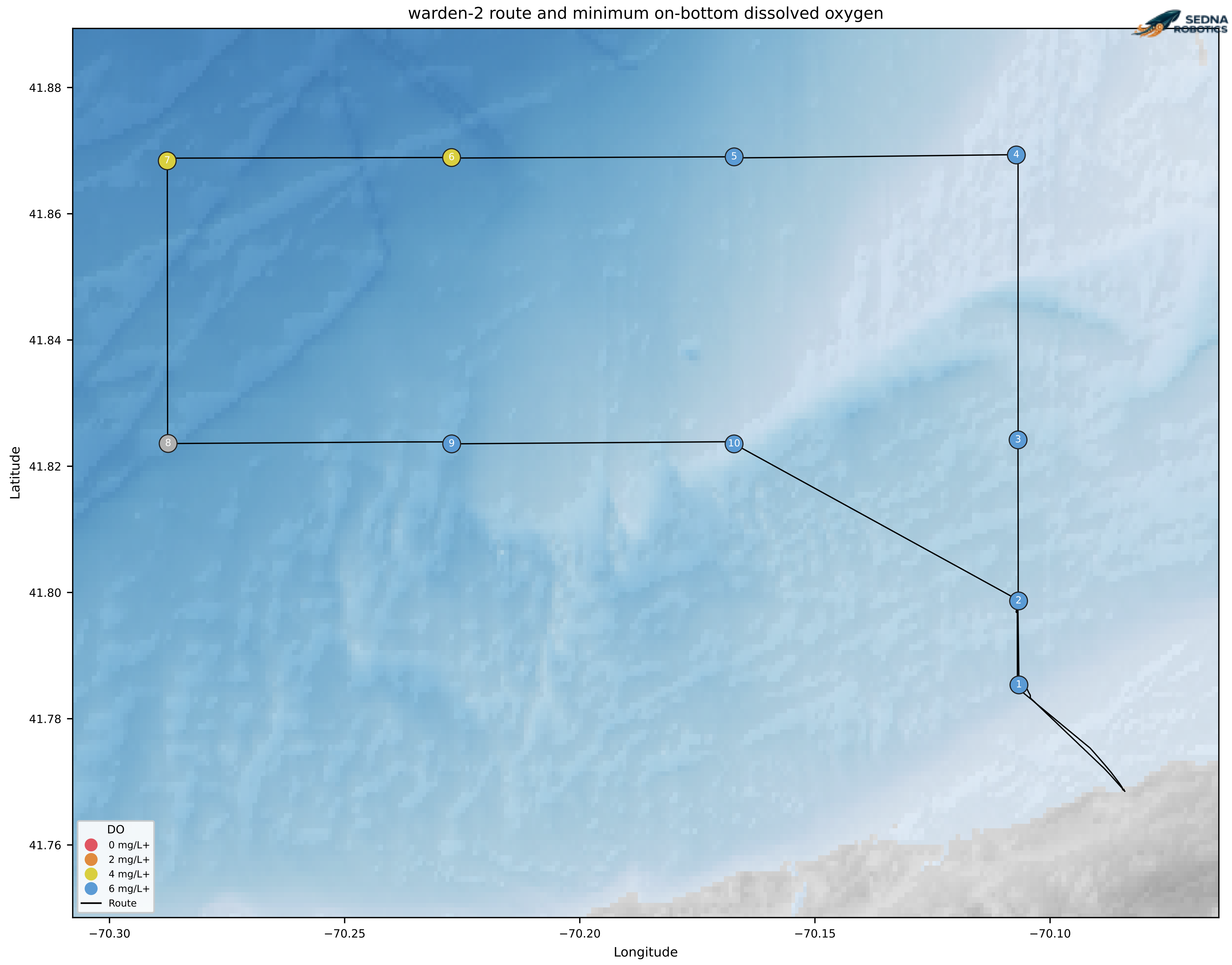
warden-2 dissolved oxygen casts — grey points tagged not-on-bottom (|winch speed| > 0.3 m/s or line length < 1.5 m)





warden-2 route and minimum on-bottom dissolved oxygen
SEDNA
ROBOTICS
41.88
41.86
41.84
41.82
41.80
41.78
41.76
Latitude
−70.30
−70.25
−70.20
−70.15
−70.10
Longitude
1
2
3
4
5
6
7
8
9
10
DO
0 mg/L+
2 mg/L+
4 mg/L+
6 mg/L+
Route

Sources, filtering and assumptions

DATA SOURCES

- Vehicle telemetry: InfluxDB Cloud bucket warden, callsign warden-2, measurements global_position_int, winch_status, profile_sample.
- Bathymetry: NOAA NCEI best-available DEM mosaic (DEM_mosaics/DEM_all ImageServer). In this area the mosaic is fed by the CUDEM 1/9 arc-second tiles ncei19_n41x75/n42x00_w070x25/w070x50_2021v1, exported at 1/3 arc-second (~10 m).
- Tides: NOAA CO-OPS predictions, station 8447241 Sesuit Harbor, East Dennis (41.7517°N, 70.1550°W), datum MLLW, metric, 6-minute interval.
- Vertical datum transform: NOAA VDatum, MLLW→NAVD88 = -1.764 m (±0.117 m) evaluated at the track centroid.

FILTERING AND PROCESSING

- global_position_int and winch_status reduced to 1 Hz server-side with aggregateWindow(every: 1s, fn: last, createEmpty: false); profile_sample kept at its native 30 s rate.
- Winch line length and speed are linearly interpolated in time onto each DO sample timestamp; gaps longer than 30 s are left null.
- A reading is tagged not on bottom when |winch speed| > 0.3 m/s or line length < 1.5 m. Tagged readings are greyed out and excluded from the station minimum shown on the chart.
- Seabed elevation is sampled bilinearly from the DEM at each vessel fix.

ASSUMPTIONS AND CAVEATS

- DEM elevations are treated as metres relative to NAVD88, the CUDEM vertical datum. bathymetry_depth_m is the negated seabed elevation; tide_adjusted_depth_m adds the predicted water level.
- Tides are predicted astronomical levels, not observed water levels: storm surge and wind setup are not represented.
- One tide station and one VDatum offset are applied across the whole survey area, so spatial variation in tidal phase, range and datum separation is ignored. Sesuit Harbor is a subordinate station, derived from a reference station rather than measured harmonics.
- Winch line length is used as the depth proxy for each reading. There is no wire angle or layback correction, so a reading's true depth is at most its line length.
- The Lowell Instruments DOT-2 logger does not report pressure (profile_sample.pressure_dbar is zero throughout), so no independent sensor depth is available as a check.
- Positions are the vessel GPS fix (global_position_int), not the position of the sensor itself. The vessel free-drifts during sensor deployment.
- Quoted uncertainty covers the VDatum transform only. DEM vertical uncertainty, tide prediction error and GPS error are not included.
- Where every reading at a station is tagged, no minimum on-bottom DO exists and the station is drawn grey.